THE CECH COMPLEX OF PUSHFORWARDS OF THE RESTRICTIONS IS A RESOLUTION OF THE O_X-MODULE

THE STACKS PROJECT AUTHORS

Editorial note. Preamble: only the retained preamble of this chapter (sources/stacks-cohomology/retained-preamble.tex) is taken, because the chapter file itself carries just an input of the project's single shared preamble on its first line. It holds the Xy-pic setup, the author's native theorem declarations and the mathematical macros, copied line for line; nothing was added, removed or redefined. This card uses neither stix nor transparent, which the host TeX Live lacks.

Editorial note. The display of the Cech differential (equation-d-cech, source lines 895-908) is printed as a context span, so that the differential formula which the counted statement names is visible in the material. Consequently this card needs no external newlabel stub at all; both local trial XeLaTeX runs log no undefined reference.

Editorial note. Original numbering: in this chapter lemmas, equations and remarks share the subsection counter, and material-only assembly locates passages by label, so the printed numbers need not agree with the original ones.

Editorial note. Uniform work/source_title across every card of this source, as section 11 of NEW_TASK_AUTHORING.md requires (the manifest field work is taken from source_title). The precise section of each claim is carried by claim_key and by statement_anchor in the registration, and the section name is repeated in the editorial notes where it matters.

Allowed standard prerequisites (editorial).

- The Cech complex of an open covering of a sheaf of modules and its coboundary formula. Same chapter, the display labelled equation-d-cech (canonical Stacks tag 01EE, source line 902, shown as context above); the counted statement names it for the differential.. Fixes the terms C^p(U,F) = prod_{i_0...i_p} (j_{i_0...i_p})_* F_{i_0...i_p} and the differential d whose exactness on stalks the proof verifies.
- Exactness of a sequence of sheaves of modules can be checked on stalks, and restriction to an open subspace commutes with taking stalks. Standard sheaf theory as used throughout the same chapter's section The Cech complex and Cech cohomology (source lines 868 onwards); the proof uses it at lines 4603-4609.. Reduces the quasi-isomorphism claim to exactness of the extended complex of stalks, which is then established by an explicit contracting homotopy.

$$d : \check{\mathcal{C}}^p(\mathcal{U}, \mathcal{F}) \longrightarrow \check{\mathcal{C}}^{p+1}(\mathcal{U}, \mathcal{F})$$

by the formula

$$d(s)_{i_0 \ldots i_{p+1}} = \sum\nolimits_{j=0}^{p+1} (-1)^j s_{i_0 \ldots \hat{i}_j \ldots i_{p+1}}|_{U_{i_0 \ldots i_{p+1}}} \tag{0.0.1}$$

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Lemma 0.1. *Let X be a ringed space. Let $\mathcal{U} : X = \bigcup_{i \in I} U_i$ be an open covering of X. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module. Denote $\mathcal{F}_{i_0 \ldots i_p}$ the restriction of $\mathcal{F}$ to $U_{i_0 \ldots i_p}$. There exists a complex $\mathfrak{C}^\bullet(\mathcal{U}, \mathcal{F})$ of $\mathcal{O}_X$-modules with*

$$\mathfrak{C}^p(\mathcal{U}, \mathcal{F}) = \prod\nolimits_{i_0 \ldots i_p} (j_{i_0 \ldots i_p})_* \mathcal{F}_{i_0 \ldots i_p}$$

and differential $d : \mathfrak{C}^p(\mathcal{U}, \mathcal{F}) \to \mathfrak{C}^{p+1}(\mathcal{U}, \mathcal{F})$ as in Equation (0.0.1). Moreover, there exists a canonical map

$$\mathcal{F} \to \mathfrak{C}^\bullet(\mathcal{U}, \mathcal{F})$$

which is a quasi-isomorphism, i.e., $\mathfrak{C}^\bullet(\mathcal{U}, \mathcal{F})$ is a resolution of $\mathcal{F}$.

Proof. We check

$$0 \to \mathcal{F} \to \mathfrak{C}^0(\mathcal{U}, \mathcal{F}) \to \mathfrak{C}^1(\mathcal{U}, \mathcal{F}) \to \ldots$$

is exact on stalks. Let $x \in X$ and choose $i_{\text{fix}} \in I$ such that $x \in U_{i_{\text{fix}}}$. Then define

$$h : \mathfrak{C}^p(\mathcal{U}, \mathcal{F})_x \to \mathfrak{C}^{p-1}(\mathcal{U}, \mathcal{F})_x$$

as follows: If $s \in \mathfrak{C}^p(\mathcal{U}, \mathcal{F})_x$, take a representative

$$\widetilde{s} \in \mathfrak{C}^p(\mathcal{U}, \mathcal{F})(V) = \prod\nolimits_{i_0 \ldots i_p} \mathcal{F}(V \cap U_{i_0} \cap \ldots \cap U_{i_p})$$

defined on some neighborhood V of x, and set

$$h(s)_{i_0 \ldots i_{p-1}} = \widetilde{s}_{i_{\text{fix}} i_0 \ldots i_{p-1}, x}.$$

By the same formula (for $p = 0$) we get a map $\mathfrak{C}^0(\mathcal{U}, \mathcal{F})_x \to \mathcal{F}_x$. We compute formally as follows:

$$\begin{aligned}
(dh + hd)(s)_{i_0 \ldots i_p} &= \sum\nolimits_{j=0}^{p} (-1)^j h(s)_{i_0 \ldots \hat{i}_j \ldots i_p} + d(s)_{i_{\text{fix}} i_0 \ldots i_p} \\
&= \sum\nolimits_{j=0}^{p} (-1)^j s_{i_{\text{fix}} i_0 \ldots \hat{i}_j \ldots i_p} + s_{i_0 \ldots i_p} + \sum\nolimits_{j=0}^{p} (-1)^{j+1} s_{i_{\text{fix}} i_0 \ldots \hat{i}_j \ldots i_p} \\
&= s_{i_0 \ldots i_p}
\end{aligned}$$

This shows h is a homotopy from the identity map of the extended complex

$$0 \to \mathcal{F}_x \to \mathfrak{C}^0(\mathcal{U}, \mathcal{F})_x \to \mathfrak{C}^1(\mathcal{U}, \mathcal{F})_x \to \ldots$$

to zero and we conclude. □